

# SHUVAM PAL

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## Education

**Dayananda Sagar College of Engineering**

**Bengaluru, India**

*B.E. Artificial Intelligence and Machine Learning*

## Experience

**Bentham AI**

*Dec 2025 – Feb 2026*

*Backend Engineering Intern: Node.js, JavaScript, Go, Docker, GCP, GitHub Actions*

*Remote*

- Shipped production automation in a small team, connecting Node.js services, Puppeteer browser workflows, operator validation, API lookups, Docker, Cloud Run, and GitHub Actions.
- Built failure-aware execution with retries, session recovery, and state validation, reducing manual filing time by **85%**.
- Worked on AI-assisted structured drafts that were validated before downstream automation, keeping generated outputs inspectable instead of prompt-only.

**Google Summer of Code**

*May 2025 – Sept 2025*

*Mentee at Invesalio: Python, VTK, wxPython, NumPy, PyTorch, ONNX*

*Remote*

- Built Python model-inference workflows inside a mature open-source app, integrating preprocessing, model execution, generated masks, and visual inspection paths.
- Reduced processing runtime by **87%** using async/parallel execution while preserving correctness on large MRI volumes.
- Debugged PyTorch/ONNX-adjacent outputs, label mapping, data transformation, and UI integration issues across unfamiliar code quickly.

## Open Source

**Contributions**

- Contributed maintainer-reviewed patches across **Kuadrant MCP Gateway**, Kyverno, IOOS, Tiled, CRIU, and VideoLAN, with emphasis on reliability, integration behavior, parser robustness, and security-validation paths.

## Projects

**Omni-SSM Voice Agent** | *Python, PyTorch, ONNX Runtime, Sherpa-ONNX, Moonshine, Hindi ASR, Piper*

- Built a local **voice-agent research loop** combining STT/TTS experiments, SSAMBA streaming CTC, Piper low-latency TTS, optional Qwen3-TTS output, and WebRTC-style loopback benchmarks.
- Worked on Moonshine Hindi ASR tuning with manifest-driven dataset loading, teacher/student distillation, partial fine-tuning scripts, and WER, first-audio latency, memory, and provider-profile checks.

**Hyoka** | *Python, FastAPI, SQLAlchemy, Alembic, PostgreSQL, OpenTelemetry, Next.js, Docker*

- Built an **AI-agent reliability control plane** with SDK/proxy/OTLP trace ingestion, evaluator registries, replay modes, failure mining, validation runs, gate decisions, promotions, and audit logs.
- Implemented worker-owned execution leases, content-addressed artifacts, signed manifests, project-scoped API keys, SQLAlchemy/Alembic models, and a Next.js dashboard for operating agent behavior safely.

**Swish Support System** | *JavaScript, React, Python, PostgreSQL/pgvector, Redis, Langfuse*

**GitHub**

- Built a hybrid AI + rules support system where LLM assessments classify messy complaint semantics while deterministic policy code controls refunds, replacements, coupons, and escalations.
- Grounded responses in order, item, evidence, customer, and policy context with PostgreSQL/pgvector, Redis state, Langfuse traces, confidence thresholds, and regression checks.

## Technical Skills

**Languages:** Python, TypeScript, JavaScript, Go, SQL

**Frameworks:** FastAPI, React, Next.js, Node.js, PyTorch, ONNX Runtime

**Tools:** PostgreSQL, Redis, Docker, GCP, GitHub Actions, OpenTelemetry, Langfuse, Firebase Auth, Git